

# Module 0: Welcome

## Study-Aid Icons

In your exercises, you might see one or more of the following icons.



### Warning

If you misconfigure a step marked with this icon, later steps might not work properly. Check the step carefully before you move forward.



### Attention

Review this step or comment carefully to save time and avoid errors.



### Information

Review information about the topic or procedure.

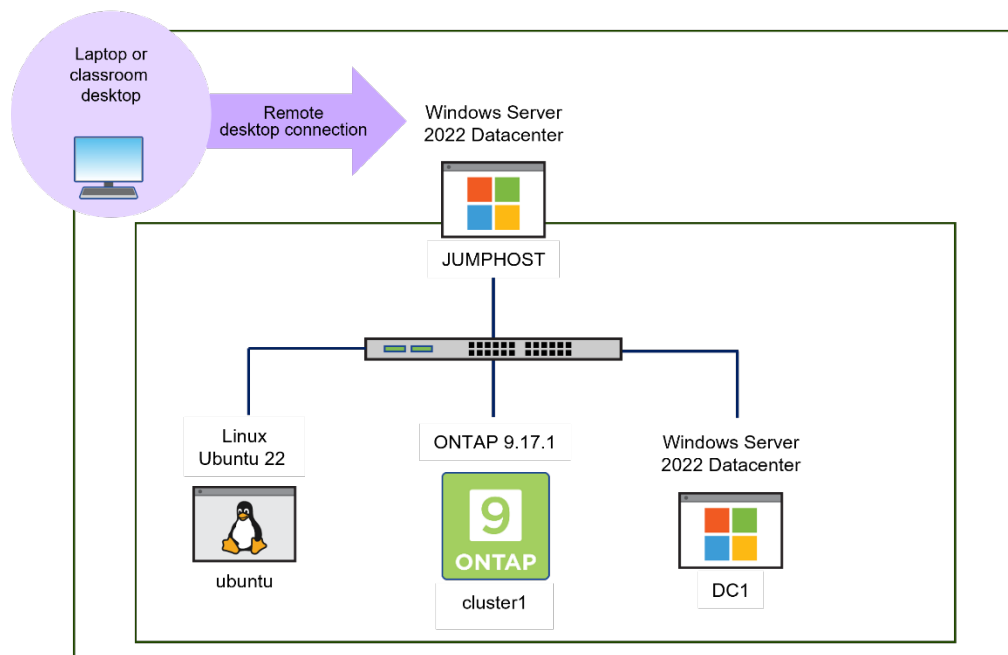
## Exercise Equipment Diagram

Your exercise environment contains one of each of the following virtual machines (VMs):

- Windows Server 2022 Datacenter (jump host)
- Windows Server 2022 Datacenter (domain controller)
- NetApp ONTAP 2-node cluster (cluster1)
- Ubuntu 22.04 LTS server system

Use the connection information that your instructor provides. When you sign in to the portal through your web browser, you are connected to a Windows Server jump host through a web-based Remote Desktop Protocol (RDP) connection. From this jump host, you connect to all other systems in your exercise environment.

Student lab environment



| System                                  | Host name           | IP address    | User name              | Password |
|-----------------------------------------|---------------------|---------------|------------------------|----------|
| Windows Server 2022 Datacenter          | JUMPHOST            | 192.168.0.5   | DEMO\Administrator     | Netapp1! |
| Ubuntu 22 Server                        | ubuntu              | 192.168.0.61  | Root (case-sensitive)  | Netapp1! |
| ONTAP cluster-management LIF (cluster1) | Cluster1            | 192.168.0.101 | admin (case-sensitive) | Netapp1! |
| node 1 (cluster1)                       | cluster1-01-console | 192.168.0.111 | admin (case-sensitive) | Netapp1! |
| node 2 (cluster1)                       | cluster1-02-console | 192.168.0.112 | admin (case-sensitive) | Netapp1! |
| Windows domain controller               | DC1                 | 192.168.0.253 | DEMO\Administrator     | Netapp1! |

# Exercise 1: Checking the Exercise Equipment

In this exercise, you familiarize yourself with your equipment, ensure that licenses are installed, and review the NetApp ONTAP System Manager UI.

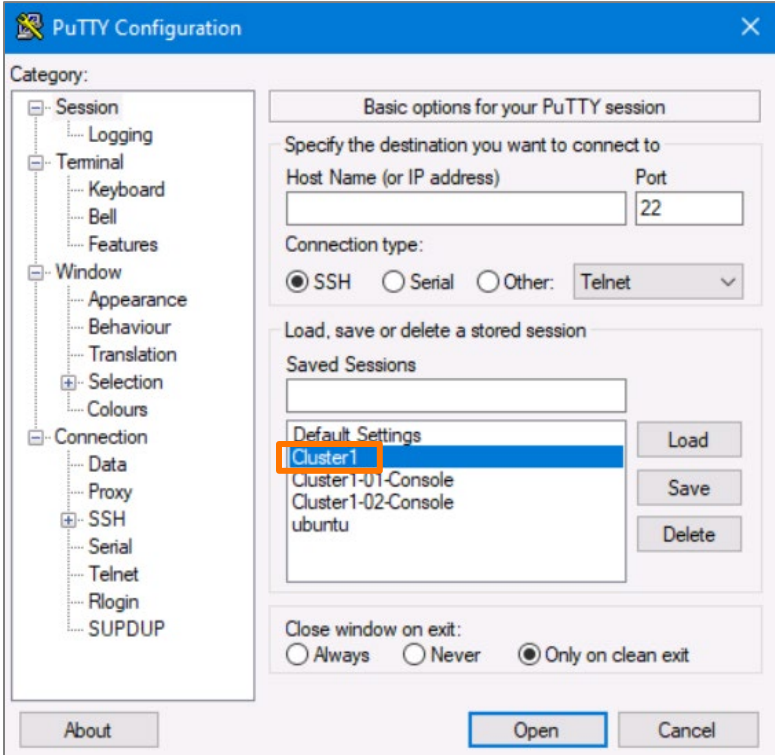
## Objectives

This exercise focuses on enabling you to do the following:

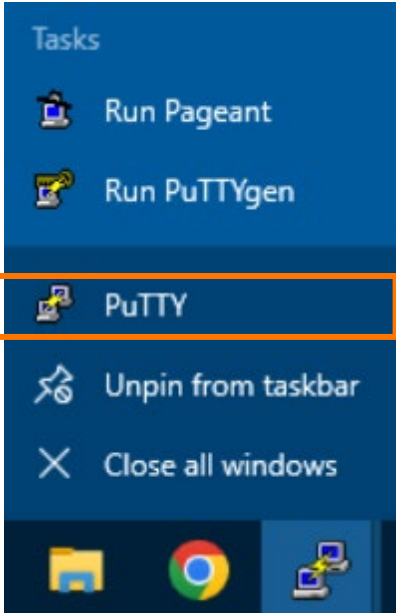
- Test connectivity to an ONTAP cluster
- Verify that required licenses are installed on the ONTAP cluster
- Log in to NetApp ONTAP System Manager and review cluster information

## Task 1: Test Connectivity to Your ONTAP Cluster

In this task, you familiarize yourself with the Windows Server desktop environment. You test connectivity to the ONTAP clusters and verify the health of the ONTAP clusters.

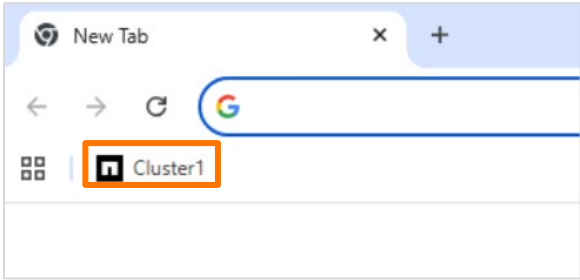
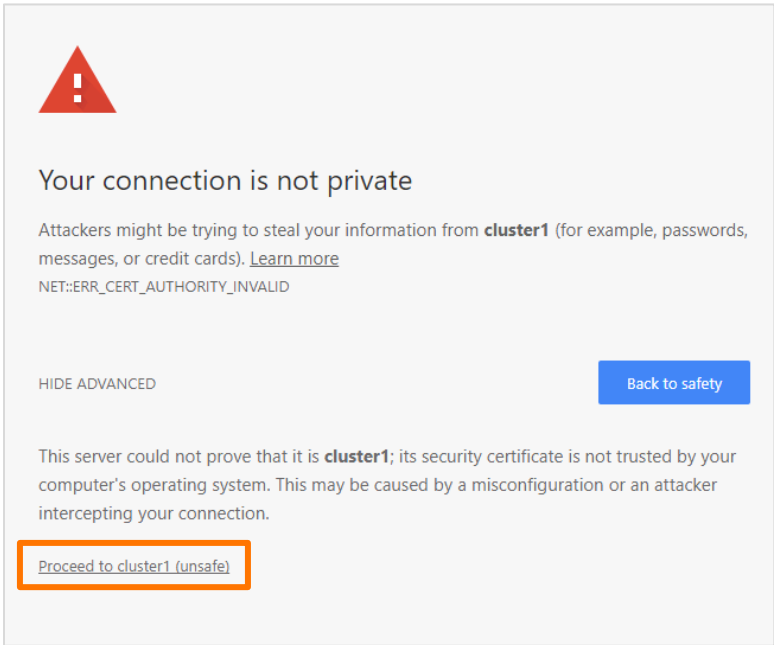
| Step | Action                                                                                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-1  | <div> To connect to the CLI of the ONTAP cluster, you use PuTTY, which is a UI for the Telnet and Secure Shell (SSH) protocols. In later exercises, you use the Google Chrome browser to connect to the ONTAP System Manager UI.</div> |
| 1-2  | <div>On the jump host desktop taskbar, locate and click the shortcut to the <b>PuTTY</b> program.</div> <div></div>                                                                                                                    |
| 1-3  | <div>In the PuTTY Configuration dialog box, double-click <b>Cluster1</b> in the Saved Sessions list.</div> <div></div>                                                                                                              |

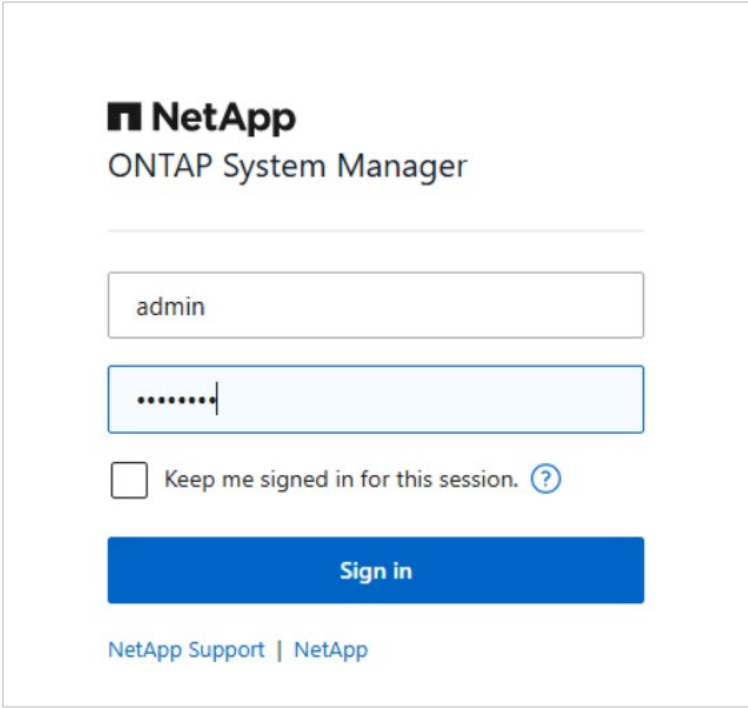
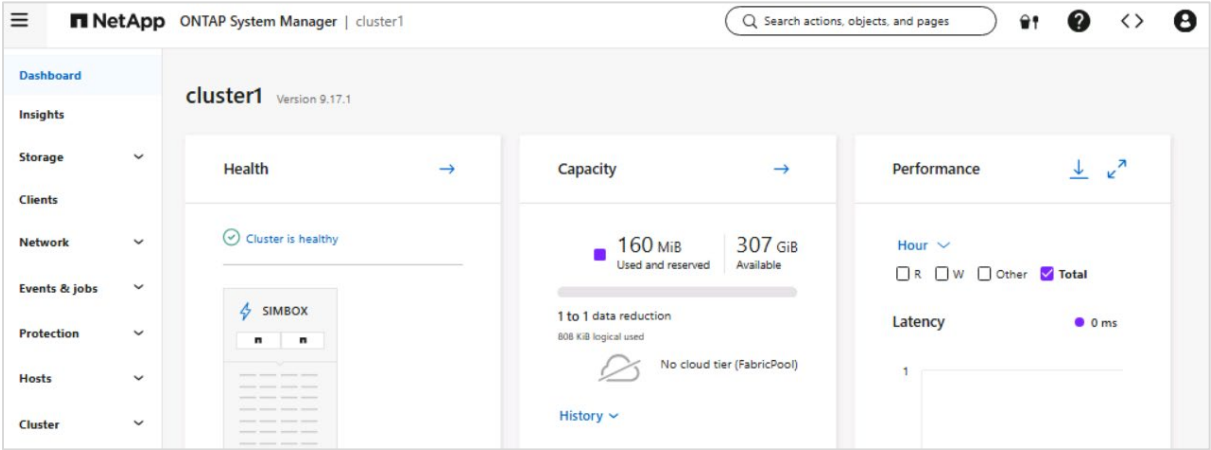
| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-4  | Ignore any security warning that might appear in the PuTTY Security Alert dialog box and click <b>Yes</b> .                                                                                                                                                                                                                                                                                                                                                                          |
| 1-5  | <p>At the ONTAP cluster login prompt, enter the following cluster1 credentials so that the ONTAP cluster CLI prompt and cursor are displayed:</p> <ul style="list-style-type: none"> <li>Login as: <b>admin</b></li> <li>Password: <b>Netapp1!</b></li> </ul> <p>Sample output:</p> <pre>login as: admin Keyboard-interactive authentication prompts from server: Password: End of keyboard-interactive prompts from server Last login time: 4/11/2025 12:25:07 cluster1::&gt;</pre> |
| 1-6  |  <p>If you have difficulty logging in to the ONTAP cluster CLI, see the table in the exercise equipment section, and verify that you are using the correct user name and password for the cluster-management LIF on cluster1. Both the user name and password are case-sensitive.</p>                                                                                                               |
| 1-7  | <p>To verify that both nodes of the ONTAP cluster are healthy and eligible, run the following command:</p> <p><b>cluster show</b></p> <p>Sample output:</p> <pre>cluster1::&gt; cluster show Node                      Health  Eligibility ----- cluster1-01              true    true cluster1-02              true    true 2 entries were displayed.</pre>                                                                                                                         |
| 1-8  |  <p>If the health or eligibility of either node is listed as <b>false</b>, inform your instructor.</p>                                                                                                                                                                                                                                                                                            |
| 1-9  | <p>To view the ONTAP software version on the cluster, run the following command:</p> <p><b>version</b></p> <p>Sample output:</p> <pre>cluster1::&gt; version NetApp Release 9.17.1: Thu Sep 04 19:55:22 UTC 2025</pre>                                                                                                                                                                                                                                                               |

| Step | Action                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-10 | <p>To open another PuTTY session, right-click the PuTTY icon in the taskbar, and then click <b>PuTTY</b>.</p>                          |
| 1-11 |  <p>You use cluster1 for all the exercises of this course.</p>                                                                         |
| 1-12 |  <p>If no CLI activity occurs for some time, the logged-in user is logged out. The default timeout value is 30 minutes.</p>           |
| 1-13 | <p>To modify the timeout intervals for CLI sessions, run the <b>system timeout modify</b> command on the cluster to set the timeout value for CLI sessions to 0 (zero).</p> <pre>system timeout modify -timeout 0</pre> |

## Task 2: Log in to ONTAP System Manager

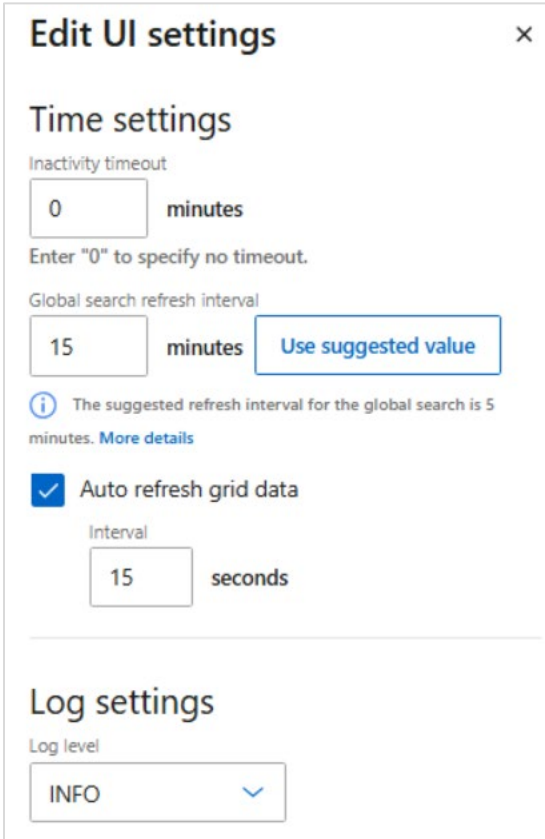
In this task, you log in to the ONTAP System Manager UI for each cluster, review the cluster information, and modify the default UI timeout interval.

| Step | Action                                                                                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-1  | <p>On the jump host desktop or the taskbar, launch the Chrome browser.</p>                                                  |
| 2-2  | <p> When you launch the browser, an ONTAP System Manager bookmark for each cluster is available in the bookmarks bar.</p>    |
| 2-3  | <p>To launch ONTAP System Manager for cluster1, click the <b>Cluster1</b> bookmark in the bookmarks bar.</p>                 |
| 2-4  | <p>Ignore any security warning that might appear, click the <b>Advanced</b> button, and proceed to the login screen.</p>  |

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-5  | <p>Log in to ONTAP System Manager for cluster1 with the following credentials:</p> <ul style="list-style-type: none"><li>▪ User name: <b>admin</b></li><li>▪ Password: <b>Netapp1!</b></li></ul> <div>The image shows the NetApp ONTAP System Manager login interface. At the top, the NetApp logo and 'ONTAP System Manager' text are displayed. Below this is a username input field containing 'admin' and a password input field with masked characters. A checkbox for 'Keep me signed in for this session.' is present, along with a help icon. A large blue 'Sign in' button is at the bottom, with 'NetApp Support   NetApp' links below it.</div>                             |
| 2-6  | <p>On the ONTAP System Manager dashboard, review information about cluster health, capacity, performance, and network configuration.</p> <div>The image shows the NetApp ONTAP System Manager dashboard for 'cluster1' (Version 9.17.1). The left sidebar contains navigation links: Dashboard, Insights, Storage, Clients, Network, Events &amp; jobs, Protection, Hosts, and Cluster. The main content area has three panels: 'Health' showing 'Cluster is healthy' with a SIMBOX icon, 'Capacity' showing '160 MiB Used and reserved' and '307 GiB Available' with a '1 to 1 data reduction' bar chart, and 'Performance' showing 'Hour' selection and 'Latency' of '0 ms'.</div> |
| 2-7  | <div>An information icon consisting of a lowercase 'i' inside a blue circle.</div> <p>If no ONTAP System Manager activity occurs for some time, the logged-in user is logged out. The default timeout value is 30 minutes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Step | Action                                                                                                                    |
|------|---------------------------------------------------------------------------------------------------------------------------|
| 2-8  | <p>To modify the timeout intervals, in the ONTAP System Manager navigation pane, select <b>Cluster &gt; Settings</b>.</p> |
| 2-9  | <p>In the Settings windows, click the <b>edit</b> (pencil) icon in the UI settings pane.</p>                              |



| Step | Action                                                                                                                                                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-10 | <p>In the Edit UI settings dialog box, review the available settings, and then change the <b>Inactivity timeout</b> field to a required value, such as 0 minutes.</p>  |
| 2-11 |  <p>Disabling automatic logout weakens security and should never be done outside of training environments.</p>                                                        |
| 2-12 | <p>Retain the default values for the other fields, and then click <b>Save</b>.</p>                                                                                                                                                                       |

**End of exercise**